

Statistical Arbitrage

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Problem 1: The OU Process

1.

$$\begin{aligned}dX_t &= \theta(\mu - X_t)dt + \sigma dW_t, \quad \theta, \sigma > 0 \\d(e^{\theta t} X_t) &= e^{\theta t} dX_t + \theta e^{\theta t} X_t dt \\d(e^{\theta t} X_t) &= e^{\theta t} (\theta(\mu - X_t)dt + \sigma dW_t) + \theta e^{\theta t} X_t dt \\d(e^{\theta t} X_t) &= \theta e^{\theta t} \mu dt + \sigma e^{\theta t} dW_t \\e^{\theta t} X_t - X_0 &= \mu(e^{\theta t} - 1) + \sigma \int_0^t e^{\theta s} dW_s \\X_t &= \mu - (X_0 - \mu)e^{-\theta t} + \sigma \int_0^t e^{-\theta(t-s)} dW_s\end{aligned}$$

2. We begin by investigating the stochastic term in the previous result:

$$\begin{aligned}X_t = \alpha + \beta, \quad \alpha &= \mu - (X_0 - \mu)e^{-\theta t}, \quad \beta = \sigma \int_0^t e^{-\theta(t-s)} dW_s \\&\sim N(0, \sigma^2 \int_0^t e^{-2\theta(t-s)} ds) \\&\sim N\left(0, \frac{\sigma^2}{2\theta} [1 - e^{-2\theta t}]\right)\end{aligned}$$

$$\begin{aligned}E(X_t) &= E\left(\mu - (X_0 - \mu)e^{-\theta t} + \sigma \int_0^t e^{-\theta(t-s)} dW_s\right) \\&= \mu - (X_0 - \mu)e^{-\theta t} + E\left(\sigma \int_0^t e^{-\theta(t-s)} dW_s\right) \\&= \mu - (X_0 - \mu)e^{-\theta t}\end{aligned}$$

$$\begin{aligned}X_t \text{Var}(X_t) &= E(X_t^2) - E(X_t)^2 \\&= E(\alpha^2 + 2\alpha\beta + \beta^2) - \alpha^2 \\&= E(\beta^2) \\&= \frac{\sigma^2}{2\theta} [1 - e^{-2\theta t}]\end{aligned}$$

This implies that the stationary distribution for $\lim_{t \rightarrow \infty} X_t$ is

$$\lim_{t \rightarrow \infty} X_t \sim N\left(\mu, \frac{\sigma^2}{2\theta}\right)$$

3.